



Global stability in large economies

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Received: 16 May 2024 / Accepted: 10 January 2025 / Published online: 20 February 2025
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Abstract

A result is obtained showing how global stability of equilibrium, under the standard tâtonnement dynamic, arises in large economies having a continuous, diffuse distribution of consumers, as well as many goods.

Keywords General equilibrium · Stability · Uniqueness

JEL Classification C620 · D510

1 Introduction

Completion of the general equilibrium program seemingly requires that one obtains a satisfactory stability result concerning an economy's equilibrium, and not just that equilibrium must of necessity exist. Hopefully, such an argument would extend to demonstrating global stability, and hence uniqueness, of the perfectly competitive equilibrium concerned. The original hope for a comprehensive result within the exchange model, using the standard tâtonnement dynamic, was first dispelled by the celebrated counterexample of Scarf (1960), and then even more so by the subsequent results of Sonnenschein, Mantel and Debreu (the *SMD* theorem; see Mas-Colell et al. 1995). Indeed, the point has been reached, today, where most economists are apparently of the attitude that, given the seeming arbitrariness of the possible economy-wide demand, there is little reason to ever expect stability of equilibrium. The ensuing severe conceptual problem that this poses for any supposed application of equilibrium analysis is, however, then generally ignored, the obvious problem being that an equilibrium that cannot be achieved must be judged irrelevant.

Debreu (1984) suggested that, given the limited implications of consumer rationality alone for aggregate demand, more substantial positive properties of an economy could be attained from assumptions on the distribution of consumer characteristics. A

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result in the classical mold is that of Knoer (1980) and Quah (1997) who show that an economy consisting of homothetic consumers with endowments distributed independently of preferences will feature global stability, even when there is a continuum of consumers. Since the basic reasoning is that there is then a representative consumer who is again homothetic, the good stability properties of the individual consumers are simply being preserved in the aggregate. Hildenbrand (1983, 1994), in contrast, emphasizes that, unlike the Knoer–Quah case, aggregate demand can be made to exhibit important properties not possessed by any one consumer’s demand. Hildenbrand goes on to show that such behavior is encouraged when a competitive economy, necessarily composed of a large number of consumers if perfect competition is to be supported, is actively modelled as having a continuum of consumers, one possessing a well-behaved distribution. In manifesting all this, Hildenbrand adopts a convenient “excess demand approach” (Dierker 1982), where matters are expressed entirely in terms of individual excess demand, instead of, as in Knoer and Quah, considering the presumably more fundamental distributions of preferences and endowments.

The current paper will then follow the approach of Hildenbrand, but further introduces into the mix the idea that not only are there many consumers but there are many goods. Though the assumption of many goods may not be as fundamental as that of many consumers for satisfaction of the perfectly competitive framework, it is nonetheless easily seen to be one necessary for any general result on global stability. Indeed, starting with just two goods, it is not overly difficult to construct exchange economies having consumers regularly distributed along a continuum, but where the resulting economies display multiple equilibria, thus lacking global stability (see Sect. 6).

Our reasoning will rest on the rationality of consumers, and so a well-behaved individual substitution effect. The absence of any need for rationality has been emphasized as a benefit when assuming particular continuous distributions of consumer behavior that, even in the possible absence of full rationality, yield attractive aggregate results (see Grandmont 1992). We, though, regard it, instead, as a virtue, or at least a consolation, that the teachings of traditional neoclassical microeconomics, and hence that of a negative substitution effect, should still have something to offer, even when concerned, as here, with the aggregate economy.

Comparing our work to Hildenbrand (1983), he essentially requires only that the aggregate substitution effect does no harm (does not have the wrong sign) and proceeds to obtain an aggregate income effect of the right, negative sign, whereas we permit the aggregate income effect to be of whatever sign it wants but argue that it eventually becomes arbitrarily small in its effect. Thus, what we require, instead, is that the substitution effect should at least maintain a certain magnitude, it then automatically being of the desired, negative sign. Our approach is then like that of Hicks (1939) (or McKenzie 2005) in the sense that the impact of individual income effects is to be small in the aggregate, though we do not argue, as does Hicks, that each individual’s income effect ought to be small in and of itself. We demonstrate, instead, that, because individual income effects are limited in scope, if you additionally suppose that consumers and their demands are many and varied, then in whatever direction one chooses to look, within a price space of large enough dimension, few of these not

necessarily small income effects, generally pointing elsewhere, will have made any appreciable contribution to the behavior of aggregate demand.¹

2 The framework

Individual excess demand over the n goods for consumer s is denoted $z(p, s) = x(p, p \cdot \omega(s), s) - \omega(s)$, for demand $x(p, y(p, s), s)$ where wealth or income is $y(p, s) = p \cdot \omega(s)$ for endowment $\omega(s)$, but where the last factor, $\omega(s)$, is typically suppressed in our notation, to emphasize nonnegative prices p . Such demand is simply assumed to always be regular, as will occur under rationality, given smooth preferences, whenever interior solutions are assured. This then also yields us the Slutsky equation for the Jacobian of excess demand $Dz(p, s)$:

$$Dz(p, s) = S(p, s) - a(p, s)(z(p, s))', \quad (1)$$

where $S(p, s)$, the substitution matrix, is negative semi-definite under rationality, and where we are letting $a(p, s) \equiv D_y x(p, y(p, s), s)$ denote the vector of the various goods' marginal propensities to consume, as appears in the second term of (1), which constitutes the income (or wealth) effect. Now, $z(p, s)$ must be homogeneous of degree zero in p , as follows from rationality, an assumption which further implies, given that goods are desirable, that individual income will necessarily all be spent (so that $p \cdot a(p, s) = 1$). Both these properties survive aggregation, so that aggregate excess demand $z(p) = \int z(p, s) d\lambda_p(s)$ will also satisfy homogeneity of degree zero, as well as obeying Walras' Law, $p \cdot z(p) = 0$.² The remaining notable property of individual demand that also survives aggregation is then that the magnitude of individual excess demand, $\|z(p, s)\|$, explodes in size as p approaches the boundary $\partial \mathbb{R}^{n+}_+ - \{0\}$, where some prices are then going to zero, and thus, aggregate $\|z(p)\|$ similarly explodes.

These properties of excess demand can be conveniently captured by expressing $z(p)$ as a vector field on the nonnegative portion of the unit sphere, $S^{n-1}_+ \equiv \{p \in \mathbb{R}^{n+}_+ \mid \|p\| = 1\}$. This is particularly appropriate when one considers the classical tâtonnement

$$\frac{dp(t)}{dt} = z(p(t)), \quad (2)$$

since vector fields represent dynamical systems, and here, excess demand $z(p)$ is the vector field representation of the tâtonnement process. The previously mentioned boundary condition serves to prevent this dynamic from ever escaping S^{n-1}_+ , and so,

¹ The important work of Vives (1987) also obtains income effects diminishing with increases in the numbers of goods, but these are individual income effects, governed by substantial assumptions on the nature of individual preferences, where it is then not necessary, or even meaningful, to speak of distributional conditions, whereas we make considerable assumptions about the distribution of consumers' demands, but then do not require any non-standard assumptions on individual characteristics. Indeed, Vives does not actually engage in a stability discussion or perform any kind of aggregate analysis, though, seemingly, his framework would have permitted him to obtain such an aggregate stability result had he so wished.

² Here λ_p is the measure whose Radon–Nikodym derivative $\frac{d\lambda_p}{d\mu}$ with respect to the reference measure μ , usually the Lebesgue one, constitutes the density function expressing the distribution of the various excess demands (see below).

solutions to the tâtonnement dynamic under study continue to exist throughout forward time.

As is well-known, the properties just listed above are sufficient to assure the existence of an equilibrium p^* for such an economy, where then $z(p^*) = 0$, though they are far from adequate to guarantee that the equilibrium be unique and globally stable, which instead requires considerably stronger assumptions. The most well-known condition for global stability is undoubtedly aggregate *WARP* (Weak Axiom of Revealed Preference). As Mas-Colell (1985) discusses, in its differential form, this requires that the Jacobian of aggregate excess demand be negative semi-definite in all directions orthogonal to this aggregate excess demand. While rational consumers satisfy individual *WARP*, it is not, in general, inherited in the aggregate from the individual, and viable conditions assuring aggregate *WARP* have proven difficult to find.

We propose instead the alternative and complementary stability condition:

$$z(p)'Dz(p)z(p) < 0, \quad z(p) \neq 0, \quad (3)$$

which we conveniently, but somewhat inaccurately, call a *negative aggregate income effect*. To the extent that it deserves this name, it is because it is requiring negativity solely in the direction of what would be that of the income effect, were we in the case of an individual consumer, just as aggregate *WARP*, often thought of as a “negative aggregate substitution effect,” is requiring negativity in all directions orthogonal to what would be that direction of the income effect.³

Expressing the Jacobian $Dz(p)$ of the aggregate $z(p)$ as the aggregate of the component Jacobians $Dz(p, s)$, inserting the individual Slutsky decompositions (1), and separating substitution and income effect components, we restate our quadratic stability condition as:

$$\begin{aligned} z(p)'Dz(p)z(p) &= z'(p)S(p)z(p) + z'(p)M(p)z(p) \\ &= z'(p)\left(\int S(p, s)d\lambda_p(s)\right)z(p) + z(p)'\left(\int M(p, s)d\lambda_p(s)\right)z(p) \\ &= z'(p)S(p)z(p) + \int -z(p)'a(p, s)z(p, s)'z(p)d\lambda_p(s) < 0, \quad z(p) \neq 0 \end{aligned} \quad (4)$$

where, in the above, we have implicitly defined the aggregate (mean) substitution effect $S(p)$ and the aggregate (mean) income effect $M(p)$. The above quadratic aggregate substitution effect always has the desired, non-positive, sign, and so interest will center on the aggregate income effect and the individual terms that make up this quadratic aggregate term.

³ Hartman (1964) discusses stability condition (3), and, in an entirely mathematical context, points out how it is complementary to the *WARP* stability condition. See Keenan and Kim (2014) for the additional argument needed to establish global stability under condition (3), when in the current setting. The Lyapunov function used is nothing other than the usual quadratic one.

3 Standing assumptions

We are going to show conditions under which economies with enough goods become globally stable. We want to bound the size of individual excess demands, but this cannot be done entirely in general, since, as already remarked, demand explodes near the boundary ∂S_+^{n-1} where some prices are then near zero. We, however, recall that the tâtonnement vector field z on S_+^{n-1} is pointing in along this boundary and so there is necessarily a neighborhood of the boundary where the dynamic is consistently contracting state space until one reaches that neighborhood's complement, which we call S_z^* .⁴ Thus, one only needs to establish our stability condition on this compact set $S_z^* \subseteq S_+^{n-1}$.

For each economy, whatever its good dimensions, at each price p such that $z(p) \neq 0$, we consider how the distribution of agents' excess demands there induces a probability measure λ_p that describes the frequency with which the different demands constitute the various vectors lying in the tangent space $T_p - \{0\}$. To incorporate our intended continuity framework, we assume that such measures are absolutely continuous with respect to the Lebesgue measure. Any such measure can be furthermore normalized to a measure $\tilde{\lambda}_p$ on the unit sphere S^{n-2} . On the other hand, we represent the uniform probability measure on that unit sphere S^{n-2} by $\tilde{\mu}_{n-2}$, often termed the sphere's Haar measure, though, since the needed dimension should always be apparent from the context, we ordinarily denote this Haar measure simply as $\tilde{\mu}$. The normalized measures are then necessarily absolutely continuous with respect to their associated Haar measure.⁵

We have:

1. *assumption 1* For any price p in S_z^* , the magnitude of individual demand is bounded by B , so $\|z(p, s)\| \leq B$, for all consumers.
2. *assumption 2* For any price p in S_z^* the magnitude of the individual marginal propensities to consume are bounded by C , so $\|a(p, s)\| \leq C$, for all consumers.
3. *assumption 3* For any price p in S_z^* , $z(p) \neq 0$, the quadratic aggregate substitution effect $S(p)$ is at least $\varepsilon > 0$ in absolute value in the direction of normalized aggregate demand, so that $-v'S(p)v > \varepsilon$ for $v = z(p)/\|z(p)\|$.
4. *assumption 4* For any price p in S_z^* , $z(p) \neq 0$, one has the density of $\tilde{\lambda}_p$ bounded by M ; that is, $\frac{d\tilde{\lambda}_p(v)}{d\tilde{\mu}} \leq M$ for any $v \in S^{n-2}$.

The most interesting and important condition is *assumption 4*, the one bounding the density of individual demands' distributional measure $\tilde{\lambda}_p$. Note that, given this condition, one has $\tilde{\lambda}_p(A) \leq M\tilde{\mu}(A)$ for any $A \subseteq S^{n-2}$, since $\tilde{\lambda}_p(A) = \int_A \frac{d\tilde{\lambda}_p}{d\tilde{\mu}} d\tilde{\mu} \leq$

⁴ For the argument, see prop. 5.1 in Dierker (1974), particularly the second and third parts of the proof. There is also prop. 5.7 in Nagata (2004), which motivates our notation, S_z^* .

⁵ Individual demands such that $z(p, s) = 0$ make no contribution to our projections below. To the extent that these instances have any mass, the expectation calculated below, being conditional on $z(p, s) \neq 0$, will simply be found to be a yet greater exaggeration of the unconditional calculation for the aggregate income effect.

$\int_A M d\tilde{\mu} = M\tilde{\mu}(A)$.⁶ That is, the weight of consumers with normalized demands lying within any given set in S^{n-2} is bounded with respect to the size of the given set. *Assumption 4* captures the idea that one supposes the distribution of demands to be *diffuse*, that is its density to be rather flat.⁷ It implies that the density of $\tilde{\lambda}_p$ must peak over that of the uniform Haar density $\tilde{\mu}$ on S^{n-2} by a factor at most M . This condition forbids discrete distributions that locate themselves entirely at a finite number of particular individual excess demands, as occurs in, say, the *SMD* theorem, where these are employed in order to generate an aggregate demand having specified, quite possibly globally unstable, behavior. In point of fact, we should clarify that what our condition really does is to prevent any continuous, but concentrated, distributions similar to these discrete distributions, since actually discrete ones are not even permitted in principle within our continuous setup. Observe that, since, in the absence of such concentrated distributions, though including, of course, the other assumptions we make, chief among them high good dimensions, we are, indeed, able to obtain a comprehensive global stability result, this dichotomy in behavior, attributable to whether there are or are not concentrated distributions, suggests that economic theorists have been singularly efficient in locating just the sorts of circumstances needed to generate arbitrary dynamics.

Now, obviously, actual economies must, indeed, have only a finite number of consumers, and even if such consumers are regarded as independent draws from some well-behaved continuous distribution, then, despite there being many goods, the luck of the draw, so to speak, may still result in an unstable economy. However, even though consumers must, admittedly, be finite in number, this must necessarily be quite a large number if perfect competition is to be viable, and the greater that number of these consumers, the more the resulting economy will resemble the one implicit in the underlying distribution from which the draws come, an economy which we argue ought to be globally stable, given that enough goods are present.

We note that all our assumptions are bounding conditions on the distribution of economic characteristics, and so, very little is being explicitly imposed on the structure of individual demand, or indeed, aggregate demand, in contrast to most other stability results in the literature. The fact that the bounds are to hold at any price is not in and of itself a binding one; the argument proceeds at the level of each price, and so the bounds can, instead, be price dependent. In any case, since S_z^* is compact, this set of price-dependent bounds can always be replaced, without loss of generality, by a uniform one. The issue is, rather, whether, with a larger number of goods, given bounds remain in force. There is of course no definitive answer to that question, except to examine the economy at hand. We remark that, unlike the other, individual consumer terms involved in the stability condition, we needn't worry about the magnitude of aggregate z overmuch, since it appears symmetrically in both the substitution and income effects, and so washes out.

⁶ This induced measure condition then yields absolute continuity of $\tilde{\lambda}_p$, in that $\tilde{\mu}(A) = 0$ implies $\tilde{\lambda}_p(A) = 0$.

⁷ Following Debreu (1972), various notions of diffuseness have been employed in the general equilibrium literature to, for instance, obtain an aggregate excess demand that is a continuous function and not just a correspondence, or to show that such results can be obtained even when preferences are not convex. See the survey of Trockel (1984).

For *assumption 3*, recall our mentioning that while it may not be the case that every individual substitution effect is negative definite, in that sense stated in *assumption 3* for the aggregate, one would still expect the property to be true for this aggregate, assuming most consumers have regular demand. Note that the assumption requires negativity in but one direction, that of aggregate demand z . As for *assumption 2*, since $p \cdot a(p, s) = 1$, the principal way for $\|a(p, s)\|$ to grow exceedingly large is when some p_i is near zero, and so, $a_i(p, s)$ can then be large while still satisfying the budget constraint. However a price p_i very close to zero would mean that p is outside S_z^* , and so need not be considered.⁸ We have already, at least implicitly, discussed how the restriction to S_z^* can assure *assumption 1*, where the condition could not be satisfied on full S_+^{n-1} .⁹

4 The result

Before proceeding, recall the high school formula

$$a \cdot b = \|a\| \|b\| \cos(\theta) \quad (5)$$

which we can think of as being applied to both $z(p) \cdot a(p, s)$ and $z(p, s) \cdot z(p)$ in the quadratic term for the individual income effect in (4), yielding what we may call $\cos(\theta_{a(p,s)})$ and $\cos(\theta_{z(p,s)})$. While the situation between $a(p, s)$ and $z(p, s)$ is symmetric, we find it more intuitive to consider the relation of the mean $z(p)$ to its cloud of surrounding $z(p, s)$ than to the cloud of $a(p, s)$. We thus crudely bound the left hand side of the individual income effect by always assuming that $|\cos(\theta_{a(p,s)})| = 1$, the worst possible case. In contrast, we analyze $\cos(\theta_{z(p,s)})$ in more detail, since, given all our other bounds to be imposed, it is the only remaining candidate for reducing the aggregate income effect to where it can be dominated by the quadratic aggregate substitution effect. The idea, of course, is that most projections of $z(p, s)$ on $z(p)$ will be small when in high-dimensional commodity space, making the relevant cosine small. Keeping with this cosine device, what we actually do is look at $|\cos(\theta_{z(p,s)})|$, which is to say, we only consider the absolute value of each individual's quadratic income effect, treating matters as if the sign of the individual quadratic income effect were positive, the wrong sign for what we really want, whether it is or not. This obviously overestimates any role in the aggregate income effect overturning stability condition (3), so if the exaggerated effect fails to do this, then the actual aggregate income effect surely will, as well. The principle being employed is that, with an increasing number of goods, the income effect will eventually become quite small, and so it will make little difference whether it is positive or negative: the substitution effect will surely dominate.

⁸ Note that (1)–(4) are assumptions, not results, and so that all we are doing is to argue that they are not unreasonable assumptions.

⁹ Whereas, here, we pose our conditions in terms of the more explicit $z(p, s)$ and $a(p, s)$, in Keenan and Kim (2022), we express the (symmetrized) individual income effect entirely in terms of its eigenvalues, and so, the discussion there centers on the single positive eigenvalue of $M(p, s) + M'(p, s)$, the only possible source of any perverse income effect.

Theorem 1 Any economy satisfying, assumptions (1)–(4), and having enough goods, will exhibit global stability.

Proof Having assembled the pieces, the argument is straightforward. We look at any given price $p \in S_z^*$, $z(p) \neq 0$, where we may then suppress p in the notation. The absolute value of each individual term, $-(z \cdot a(s))(z(s) \cdot z)$, in (3) can be written as.

$$\begin{aligned} & \|z\| \|a(s)\| |\cos(\theta_{a(s)})| \|z(s)\| |\cos(\theta_{z(s)})| \|z\| \\ & \leq \|z\|^2 BC |\cos(\theta_{z(s)})|, \end{aligned} \quad (6)$$

where we use *assumption 1*, *assumption 2*, and the property that $|\cos(x)| \leq 1$ always. Note that integrating the last term of (6) over S^{n-2} with respect to the natural measure $\tilde{\mu}$ will yield

$$\|z\|^2 BC \int_{S^{n-2}} |\cos(\theta_{z(s)})| d\tilde{\mu} = \frac{1}{\sqrt{n-1}} \|z\|^2 BC \quad (7)$$

by a well-known formula for the expected length of the projection of a uniformly random vector in the unit sphere onto a 1-dimensional subspace (such as its first coordinate), this being what the cosine's absolute value amounts to in the present circumstances. This fact, along with its proof, is widely presented in typical developments of the Johnson–Lindenstrauss theorem, a result of great importance to data science (see, for instance, Vershynin 2018, p. 111). What we now want is, rather, the expectation with respect to our individual excess demand measure $\tilde{\lambda}$, which gives us, instead:

$$\begin{aligned} & \|z\|^2 BC \int_{S^{n-2}} |\cos(\theta_{z(s)})| d\tilde{\lambda} = \|z\|^2 BC \int_{S^{n-2}} |\cos(\theta_{z(s)})| \frac{d\tilde{\lambda}}{d\tilde{\mu}} d\tilde{\mu} \\ & \leq \|z\|^2 BC \int_{S^{n-2}} |\cos(\theta_{z(s)})| M d\tilde{\mu} = \frac{1}{\sqrt{n-1}} \|z\|^2 BCM, \end{aligned} \quad (8)$$

by *assumption 4*. To complete the proof, observe that, if n is large enough, then this term is less than the quadratic aggregate substitution effect, whose minimal size is specified by *assumption 3*. Then, since the former term is the collection of all the magnitudes of the quadratic individual income effects, treated as if they are positive, whether or not they are, it is at least as positive as the actual quadratic aggregate income effect, and so its dominance by the quadratic aggregate substitution implies that the latter also dominates the actual quadratic aggregate income effect, thus assuring condition (4), and thus global stability, once this argument is applied for all $p \in S_z^*$, $z(p) \neq 0$.

5 Generalization

We introduce a strengthening of the assumption on the substitution effect, requiring full negative definiteness of the normalized aggregate substitution effect:

assumption 3' For any price p in S_z^* , $z(p) \neq 0$, the quadratic aggregate substitution effect $S(p)$ always attains at least $\varepsilon > 0$ in absolute value among all candidate unit vectors, so that $-v'S(p)v > \varepsilon$ for all $v \in S^{n-1}$ such that $v \in T_p$.

Corollary 1 *Any economy having enough goods and satisfying assumptions 1, 2, 3', and 4 will have the Jacobian of aggregate demand $Dz(p)$ being negative definite on $T_p - \{0\}$, throughout S_z^* .*

Proof Since the argument of the theorem depends only on the one-dimensional nature of $z(p)$, it can be repeated for any vector $v \neq 0$, to give full negative definiteness over $T_p - \{0\}$ (the generalized law of demand).

While we have stated the result as a structural one on aggregate demand z , the main interest for us is that, as is well-known in the classical literature, it allows one to conclude global stability for a significantly larger class of tâtonnement processes than just the simple, classical one studied here. This property, though, should also prove useful for obtaining comparative statics results (see prop. 17.G.2 in Mas-Colell et al. 1995 for a local version). We note that this result also leads to local stability being established when its reasoning is applied to just the Jacobian of the equilibrium $z(p^*) = 0$, whereas using the weaker stability reasoning surrounding theorem 1, one would need to establish that stability condition over some neighborhood of p^* , though one possibly quite small.¹⁰ It is not clear to us how the reasoning for negative definiteness would go without first establishing a result for projections onto subspaces of lower, fixed dimensions, as done in the proof of our main theorem.

6 Example

Using analytical means to verify the sufficiency of *assumptions (1)–(4)* in assuring global stability for some interesting class of continuum economies not already known to be globally stable would be a considerable task. This is particularly so since, to illustrate the ideas, what one would really like is a sequence of economies, differing mostly in the number of commodities, where those with fewer goods are globally unstable, whereas global stability emerges with a large enough number of goods. We have, indeed, illustrated this idea with an explicit sequence of *CES* (constant elasticity of substitution) economies, where consumers' preference and endowment parameters are distributed in a well-behaved, continuous manner. Rather than trying to analyze the underlying continuum economies directly, though, we contented ourselves with making independent and identical draws from the given continuous distribution of the consumers, and then numerically studying the stability behavior of the resulting finite economies.

The numerical results show that, as the number of consumer draws increase, the asymptotic dynamic behavior of the various resulting finite economies, for any given number of goods, approach one another, that then being the dynamic behavior of the underlying continuum economy having that number of goods. The preference and endowment distributions were chosen to be symmetric with respect to the given number of goods, so that this number is the sole factor distinguishing the various continuum economies. For the chosen parameters, we explicitly looked at the cases of 2, 3, 5, and 10 goods. While parameters were chosen so that the economy with only 2 goods would be unstable, global stability already emerges by 5 (though not 3) goods, and, so,

¹⁰ In the reasoning for global stability, the non-small region S_z^* serves as this neighborhood.

unsurprisingly, continues with 10 goods. Searching among 25,000 randomly chosen price combinations not too near the boundary of S_+^{n-1} , our stability condition was always satisfied in the cases of 5 and 10 goods, and, indeed, the normalized Jacobian was always negative definite. Thus, for 5 or 10 goods, while the resulting bounds used in *assumptions (1)–(4)* are adequate to assure global stability given enough goods, having only 5 or 10 turns out to be sufficient, whereas having only 2 or 3 goods is inadequate for their associated bounds. The relatively few goods needed here for stability can, in part, be attributed to the perfect symmetry of the distributions in regard to commodities, this having been instated, as said, in order that there can be a sequence of economies differing only in that number of goods.¹¹

7 Conclusion

We have looked for global stability of the classical tâtonnement process, using no assumptions on consumers' demands other than that they feature the regularity common to standard microeconomic reasoning. In this neoclassical tradition, our reasoning has proceeded by the well-worn approach of comparing substitution to income effects, though, in a less conventional fashion, we have also emphasized the role of many goods. Since we also require many consumers, a condition needed in any case to have perfect competition, we may speak of studying large economies.

We have observed that you don't need the law of demand to hold in all directions (i.e., a negative definite Jacobian) in order to get stability, nor do you need it in all but one direction (i.e., *WARP* in the aggregate): you only need it in that one remaining direction (being our stability condition), the direction of aggregate demand. Now, there is no particular reason for this stability condition to hold for a single individual, but Hildenbrand has emphasized that the economy can exhibit behavior which no consumer does. He has emphasized, as well, the constructive role of having continuous, well-behaved distributions of consumer characteristics. While we do impose conditions of good behavior on our continuous distributions, it seems to us that they are quite mild in nature, except perhaps for the condition that the distribution of demand be diffuse. This is needed to prevent demands collecting at just the wrong places, in which case, as is well-known, stability can no longer be expected.

The most insightful argument we can offer to explain what is happening is to observe that individual income effects act in an entirely wrong, positive, fashion only in one direction, and when goods are many and the consumers are diffuse, few consumers will have their direction of positive income effect lying anywhere near the direction observed to be the only one of consequence for global stability, the direction of aggregate demand. This, to us, is the elegant explanation, but it must be admitted, that, just to get global stability, we have actually employed much less involved arguments. We

¹¹ The general parametric setup used in the numerical calculations is discussed in more detail in Keenan and Kim (2022). The simulations referred to here differ from those there only in that the decay parameter in the exponential distribution describing the economy-wide density of the various consumers' elasticity of substitution was then given the yet more extreme value of 12, rather than the previous value 3, in order to get instability when having 2 (or 3) commodities. This then makes goods, for typical draws, highly complementary, much as in the three-good counterexample of Scarf (1960). The resulting scenario is described in more detail in Keenan and Kim (2023).

have ignored whether a person's income effect falls on aggregate demand in a positive fashion or whether it falls in a negative one, merely observing that, with enough goods, the effect will be small for most of the diffused consumers, whatever the sign may be, and so, one can be confident that the always well-behaved aggregate substitution effect will have its way, assuring global stability of equilibrium in large economies.

Funding Open access funding provided by CY Cergy Paris Université.

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